

1 Setup of Composition

Usage

You use the "composition" application to control planning, monitoring and execution of the melting process.

The sections that follow describe how to configure composition functions.

General

Composition always uses unique units of weight ("kg", "KG", "t"). Consequently, the target quantity unit of charging orders is identical to the unit of weight of input materials in the batch stock.

Basic configuration

Please proceed as follows if the composition function is installed on an already existing HYDRA 8 system:

1. Enable the patch dbp_mpl_composition as follows:
 - a. UNIX systems (run at server prompt within the HYDRA directory): *hydscri.out db_sql/dbp_mpl_composition.hsc*
 - b. Windows systems (run in a DOS window within the HYDRA directory): *hydscri.exe db_sql/dbp_mpl_composition.hsc*
2. Check the patch output
3. Save the existing dialog configuration:
 - a. UNIX systems (run at the server prompt in the HYDRA directory): *hydlgcfg.out DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg*
 - b. Windows systems (run in a DOS window within the HYDRA directory): *hydlgcfg.exe DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg*
4. Now load the new dialog configurations by the following command:
 - a. UNIX systems:
hymw.out -u9999 -b db_sql/aip_gat.dlg
 - b. Windows systems:
hymw.exe -u9999 -b db_sql/aip_gat.dlg

Now the dialogs C_CHPF, C_CHCF, C_CHTS, C_CHCA are imported **as template** (with type "DEF" and dialog user "999") or, if necessary, existing dialogs are updated.

5. Copy the dialogs C_CHPF, C_CHCF, C_CHTS, C_CHCA from the template (type "DEF" with dialog user "999") to type "DEF" and dialog user "0" using the MOC application: system administration --> terminals --> dynamic dialogs (transaction code ddconf). To do so, switch to the HYDRA Professional Mode.
6. Enable the new dynamic dialogs:

- a. UNIX systems:

hydialog.scr AIPTNR 0

- b. Windows systems:

sh.exe hydialog.scr AIPTNR 0

Please note: This command enables the default dialogs. Provided that dialogs specific to the terminal are used on the system, they need to be modified by an MPDV consultant.

Basic parameter settings

Enable the option "Automatic generation of batch no. when creating batches" in the MPL tab of the basic settings in MOC (system administration --> system settings --> basic settings).

Configuration: INI configuration

The below-mentioned settings entered in the [INI configuration](#) "MPL" within the section "COMPOSITION" affect calculation of the average material price for composition within the material master:

Key	Value	Comment
FIELD_SUM_CONSUMPTION	ist_lst_01	DB field for sum of consumption in order status (Sum of consumption)
FIELD_SUM_TOTAL_PRICE	ist_lst_02	DB field for sum of the total price in order status (Total price for all component entries)
CALCAVG_BEGIN_DAYS	10	Begin date (= today - <value>) for calculation of the average price
CALCAVG_END_DAYS	5	End date (= today - <value>) for calculation of the average price

The following conditions need to apply:

- CALCAVG_BEGIN_DAYS >= CALCAVG_END_DAYS >= 0
- Valid columns of the order status (table auftrag_status) have to be indicated for the fields FIELD_SUM_CONSUMPTION and FIELD_SUM_TOTAL_PRICE.

Please note:

- The configured columns (ist_lst_01 and ist_lst_02) are calculated when finishing the charging order.
- The ini configuration is not enabled by default.

Scheduler configuration

Enter the following values in MOC System administration → System settings → [Scheduler](#) in order to calculate average values or material costs and to complete inspection requests for finished orders:

Type	S – Standard
Alterable	Yes
Type	F - Fixed
Visible	Visible
Active	☒ Active
HYDRA user	0
Command	sh.exe hycompupd.scr (Windows) hycompupd.scr (Unix)
Comment	Composition tool
Fixed point in time	Hour: 0 Minute: 30

Configuration of units

Create the [unit](#) "%" (if it does not yet exist) to be able to use it for composition recipes.

Activation at the machine

Define the previous and subsequent material buffer in the [machine master](#) --> workplace configuration --> MPL:

Parameter name	Value
Workplace type	M = melting aggregate / furnace
Maximum capacity	Specification of the maximum load capacity of the furnace in kg
Batch management	N

Parameter name	Value
Preceding material buffer	Material buffer of the type C (casting buffer) This material buffer is the collection point in front of the furnace. Input materials are gathered on this material buffer. This material buffer needs to be assigned to <u>exactly</u> one machine (unique assignment). It is not allowed to assign it to several machines. But the system does not prevent it.
Subsequent material buffer	Material buffer of the type C (casting buffer) This material buffer represents the contents of the furnace. Once the charging process has been confirmed (i.e. the materials gathered on the collection point have been put into the furnace), this buffer includes a batch of the output material. This material buffer needs to be assigned to <u>exactly</u> one machine (unique assignment). It is not allowed to assign it to several machines. But the system does not prevent it.

Activation at the material buffer

Configure one or several [material buffers](#) as casting buffers that include the anonymous (input) material (see material type):

Parameter name	Value
Type	C =Casting buffer

Activation at the material type

Configure a [material type](#) for the batches included in the casting buffers:

Parameter name	Value
Input batch processing: inventory management	A = Anonymous

Definition of input materials

Create all materials that you would like to use as input material for composition in the [material master](#).

Parameter name	Value
Material number	"Material number"
Input material for composition: Input material	☒

Activation at order type

Configure an [order type](#) for charging orders. Scheduling must not release the order.

Parameter name	Value
Order type	e.g. "CHRG"
Options/order type for composition	C = charging order
Planning/scheduling without implicit release of orders	☒

Configure an order type for melting orders.

Parameter name	Value
Order type	e.g. "MELT"
Options/order type for composition	M = melting order

Please note that the generation of inspection requirements is pre-configured for the order type "CHRG" if the order status switches from "N" (not free) to "V" (prepared). Customizing services by MPDV are required if another order type is used (table: ade_aart_cbereich).

Configuration of number ranges

Create a [number range](#) for charging orders (object "AUNR", key "AART" and value "CHRG" or the relevant order type for charging orders). This number range is required to be able to create charging orders using the function "generate charging order" in the MOC application "composition".

Example

Create number range

Configuration

Main page

Object: AUNR

Key: AART

Value: CHRG

Type: V

Assignment code: NUM

Prefix:

Range from: 100000

Range to: 199999

Current value: 100000

Configuration of order statuses

Configure the [order status](#) "N" as the initial order status for the order type that has been defined as charging order by the order type for composition (e.g. "CHRG").

Parameter name	Value
Order type	e.g. "CHRG"
Status	"N"
Data collection / control	S = None
Options / Initial status for the creation	☒

Configuration of permitted input material

Define for each output material, all [permitted input materials](#) that may be used for producing the melt of the output material. Consequently, the composition function allows for data to be restricted to the permitted materials, which simplifies the selection process.

Configuration of characteristics

Create a characteristic for each chemical element (e.g. Al, Cu, Si, Mn) that is to be considered in composition. The characteristic number and characteristic designation are only relevant to composition. The characteristics are created in the [composition recipe](#).

Configuration of composition recipe (target analysis)

Create a [composition recipe](#) (target analysis) specifying the components (see characteristics) of the material in percent (specifications: upper tolerance limit, target value, lower tolerance limit) for each input material. The target analysis defines the expected composition/make-up of an input material.

Configuration of restricted material usage

Formulas have to be defined and assigned to the recipe/material to map specific conditions (e.g. share of silicon at least 3x share of iron --> $Fe \geq Si * 3$).

This is performed within the application [material usage restrictions](#).

AIP configuration

Define the following buttons for [AIP](#): "perform charging", "confirm charging", "take sample" and "cast" in the file ctaipbut.ini.

```
[ANR-ALL-Page1]
...
6=C_CHPF,R,Perform charging
7=C_CHCF,R,Confirm charging
8=C_CHTS,R,Take sample
9=C_CHCA,R,Cast
```

The layout of the charging list is configured within the section [charge list] of the layout configuration file ctaipplay.ini:

```
[charge list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

MAT_VIS=C25,80,L
MATBEZ_VIS=C30,115,L
SOLL_MENGE_VIS=N12.3,78,R,target quantity
EINH=C3,40,L
RES_MENGE_VIS=N12.3,78,R,remaining quantity
VERBR_MENGE_VIS=N12.3,78,R,delivered quantity
MATPUF=C30,80,L
MATPUF_MENGE=N12.3,78,R,remaining quantity
EMAT_MENGE_VIS=N12.3,78,R,input buffer
```